More SQL

MBUA 512

Class 4 – More SQL

SQL = Structured Query Language, "es-queue-el" or "sequel". With Michael Winikoff.

- SELECT calculations - and DISTINCT
- WHERE conditions
- ORDER BY (DESC)
- LIMIT n
- GROUP BY and aggregates (min, max, avg, count, sum)

How to write SQL queries & tips

Based on slides by Professor Markus Luczak-Roesch.

Recap: SQL commands

- `CREATE TABLE` and `ALTER TABLE` – define the structure of a table (its *schema*)
- `INSERT INTO`, `DELETE FROM`, `UPDATE` – change the data in the table
- `SELECT` – query data

(there are a lot more than these)

> Important: make sure you select `MySQL` (not `Trino`) and your own workspace (`user_[username]`)

SELECT calculations

SELECT can include calculations:

```
SELECT concat(firstName, ' ', lastName) -- contact joins strings together
/*
current_date is the current date
datediff calculates the difference between two dates in days
floor rounds a number
*/
SELECT floor(datediff(current_date,birthdate)/365)
```

SELECT named columns

Can use `AS name` to give a column a name

```
SELECT concat(firstName, ' ', lastName) AS Name
SELECT floor(datediff(current_date,birthdate)/365) AS Age
```

SELECT DISTINCT

- Just add `DISTINCT` to remove duplicates *rows* from the result
- Example: select all pilots who fly for AirNZ

WHERE conditions

- IS NULL
- IN: `month IN (1,2,3)`
- LIKE: `email LIKE '%@%.%'` - `%` means "any string of characters" and `_` means any single character
- numerical conditions (< > = >= !=)
- date conditions (< = between): `date BETWEEN d1 AND d2`
- logical operators (AND, OR, NOT)
- (not) EXISTS (next slide)

EXISTS

- A WHERE condition is applied to each row
- An EXISTS condition allows us to check a condition for a given row against a whole table
- Example: does someone else have the same birthday?

```
SELECT personID, firstName, lastName, birthdate FROM people p
WHERE EXISTS (select * from people p2 where p.birthdate=p2.birthdate
  AND p.personID != p2.personID);
```

Note use of names to refer to row in the subquery.

UNION, INTERSECT

ORDER BY (DESC)

LIMIT n

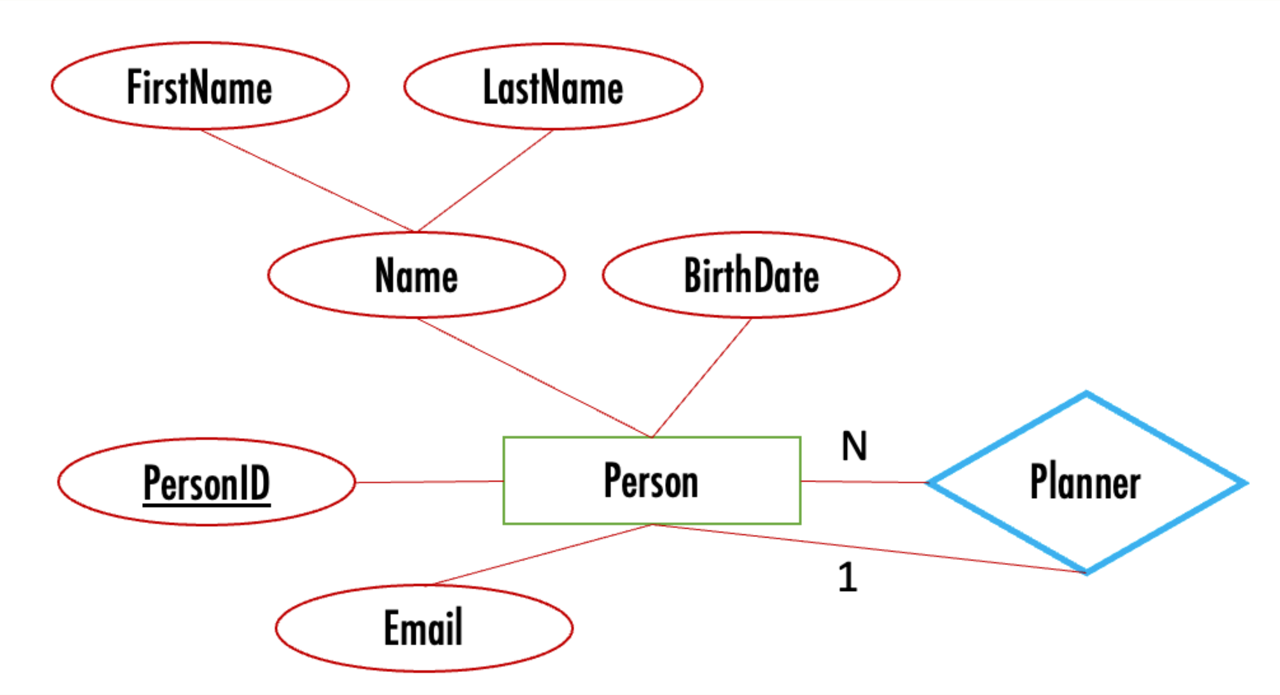
GROUP BY

Aggregates (min, max, avg, count, sum)

Running Example: birthday

- Scenario: send a message to people on their birthday, and remind their designated party planner to plan a party earlier
- Data: name (first, last), birthdate, email address, and designated party planner

Use this file to create the database.



Some queries to write (1/2)

1. Find people whose birthday is today so we can email them (combine first and last names)
2. Find people with a special birthday (e.g. 100) coming up (hint: need to calculate age, use FLOOR)
3. Find people with a birthday in January, February or March so we can email their planner (hint: use IN)

Some queries to write (2/2)

administrative tasks:

4. Find people with missing or invalid email addresses (hint: use NULL AND ... NOT LIKE)
5. Find where two (or more) people have the same email address (hint: use EXISTS)
6. Find people who are not planners for anyone else and do not have a birthday (hint: use EXISTS)

analysis task:

7. how many people have a birthday in each month? (hint: use GROUP BY and EXTRACT(MONTH from ...))

Tips for writing SQL

- Build it up in pieces, testing each piece (`*` can be useful)
- Think about:
  - What information you need to see in the result (which columns ...) ... values? Calculations? Aggregate calculations (e.g. SUM)?
  - What tables they come from
  - How you need to JOIN the tables
  - What constraints to use ... and how to express them
    - WHERE if based on row
    - WHERE ... AND exists – if not

More tips

- `' '` (single quotes), not `"` (double quote)
- Distinct – the whole row needs to be distinct, so sometimes select fewer columns
- Group By ... Having ... - the ”Having” goes just after the “Group By ...”
- NULL is not the same as the empty string
- Too many rows in result? Check JOIN conditions (or using “NATURAL JOIN” - which is not the same as “JOIN”)
- No rows in result? Check conditions, and perhaps remove some
- Using exists – make sure there is also an “AS ...” (so can refer to parent query in sub-query)

Activities

Activities – More SQL

1. Go to learn.datascie.nz and login if needed
2. Select `MySQL` (not `Trino`) and your own workspace (`user_[username]`)
3. If needed, create the flight DB using this file

Write queries to answer the following questions

TO COMPLETE

Activity – ...

~ 10 minutes, in groups